

Probability 1 – Exercises

Tutorial no. 12

4th Dec 2025

12.1 Let X and Y be two absolutely continuous random variables. Let X have a marginal PDF of $f_X(x) = \lambda^2 x \exp(-\lambda x)$ if $x \geq 0$, and 0 otherwise. Conditional on $X = x$, let Y be uniform on the interval $[0, x]$. What is:

- (a) the joint PDF of (X, Y) ?
- (b) the marginal PDF of Y , and $\mathbb{E}(Y)$?
- (c) $\mathbb{E}(X | Y = y)$ and $\mathbb{E}(Y | X = x)$?
- (d) $\mathbb{E}(X)$? (Hint: use previous results)
- (e) $\text{Cov}(X, Y)$?

12.2 Let $X_1, \dots, X_n$ be independent and identically distributed random variables (iid). What is $\mathbb{E}(X_1 | X_1 + \dots + X_n)$?

HW 12.3 (2 points) Let X and Y be i.i.d. (independent and identically distributed) positive random variables.

$$\mathbb{E} \left(\frac{X}{X + Y} \right) = ?$$

12.4 Let Y be a normal variable with mean μ and variance 1, and the conditional distribution of X given $Y = y$ is normal variable with mean y and variance σ^2 . What is the conditional distribution of Y given $X = x$?

Interpretation: Y is the original signal, X is the observed noisy version of the signal (i.e., the additive noise was an independent normal variable with mean zero and variance σ^2) and after detecting that value of the noisy version, we want to infer the original signal.

HW 12.5 (3 points) We take a PQ line segment with length 2 and break it into two parts, at a uniformly chosen point R . Let X denote the length of the PR line segment. The other, $2 - X$ long line segment is broken again, at a uniformly chosen point S . Let Y denote the length of the RS line segment.

- (a) What is the joint PDF of (X, Y) ?
- (b) Find $\mathbb{E}(Y | X = x)$.
- (c) Calculate $\text{Cov}(X, Y)$.

♣ 12.6 (3 points) Let X, Y and Z be independent random variables. Let the CDF of X and Y be $F(x)$ and $G(y)$, respectively. Let Z be a Bernoulli random variable with parameter p . Determine the CDF of the following distributions:

- (a) $T := ZX + (1 - Z)Y$,
- (b) $U := ZX + (1 - Z) \max\{X, Y\}$,
- (b) $U := ZX + (1 - Z) \min\{X, Y\}$.

12.7 We roll a die. If the result is i , then we choose Y uniformly on $[0, i]$. What is $\mathbb{E}(Y)$ and $\mathbb{D}^2(Y)$?

12.8 The amount of time until a lightbulb breaks follows an exponential distribution with 9 months expected time. If the lightbulb breaks, I immediately replace it with the same kind of lightbulb. I use my lamp for T years, where T is a number uniformly chosen between 0 and 9 years. Let Y be the number of times I have to change the lightbulb during this time. $\mathbb{E}(Y) = ?$ and $\mathbb{D}^2(Y) = ?$

12.9 Let U_1 and U_2 be two independent random variables with uniform distribution on $[0, 1]$. Show that if

$$\begin{aligned} X &= \sqrt{-2 \ln U_1} \cdot \cos(2\pi U_2) \\ Y &= \sqrt{-2 \ln U_1} \cdot \sin(2\pi U_2) \end{aligned}$$

Then (X, Y) have a joint Normal distribution.

HW 12.10 (3 points) Let X and Y be the two coordinates of a point that we choose randomly on the disc with radius 1 around the origin. ($f(x, y) = \frac{1}{\pi}$ if $x^2 + y^2 \leq 1$.) What is the joint density function of $R = \sqrt{X^2 + Y^2}$ and $\Theta = \arctg(Y/X)$?

12.11 Let $X, Y \sim \text{Exp}(\lambda)$ be independent. What is the joint density of $U := X + Y$ and $V := \frac{X}{X+Y}$? Show that U and V are independent.

HW 12.12 (2 points) Let $X_1, \dots, X_n, X_{n+1}$ be independent, standard normal random variables, and let

$$Y = \frac{X_1 + \dots + X_n}{n}.$$

(a) What is the covariance matrix of (X_1, Y) ?

(b) Find $\mathbb{P}(|Y| \leq |X_{n+1}|)$.

12.13 Two rival research groups are trying to estimate the mass of a particle. The first group performs k measurements, while the second group performs l measurements. Both research groups estimate the mass of the particle by taking the empirical average of their respective measurements. The true mass of the particle is μ , and the (noisy) measurements have normal distribution with mean μ and variance of σ^2 , independently of each other. What is the probability that the estimate of the first research group is better than that of the second research group?

12.14 Let X be a standard normal distribution, and I independent from X , with $\mathbb{P}(I = 1) = \mathbb{P}(I = 0) = \frac{1}{2}$. Define Y as

$$Y := \begin{cases} X, & \text{if } I = 1 \\ -X, & \text{if } I = 0. \end{cases}$$

(a) Show that Y is standard normal.

(b) Are I and Y independent?

(c) Are X and Y independent?

(d) Show that $\text{Cov}(X, Y) = 0$.